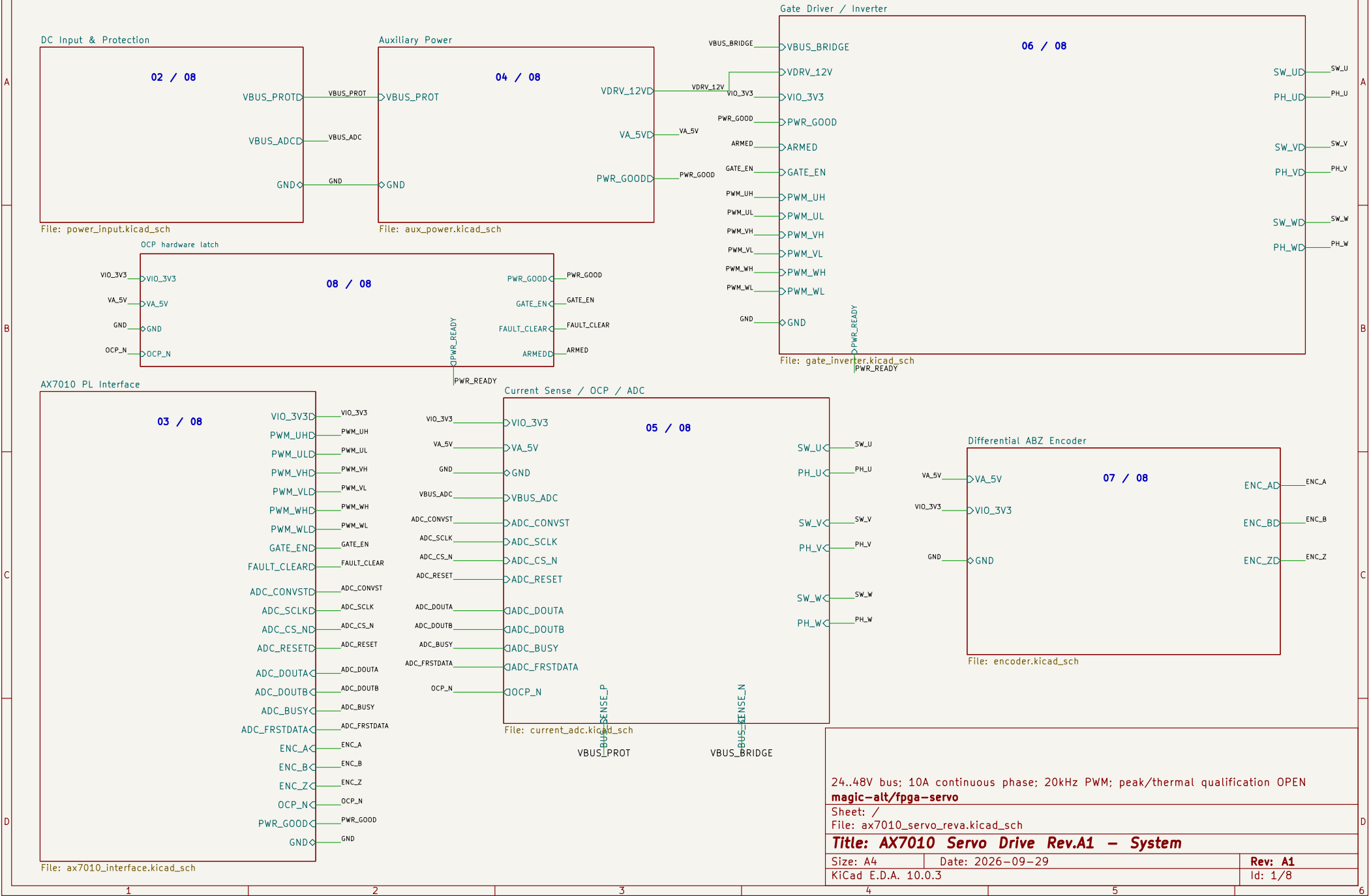
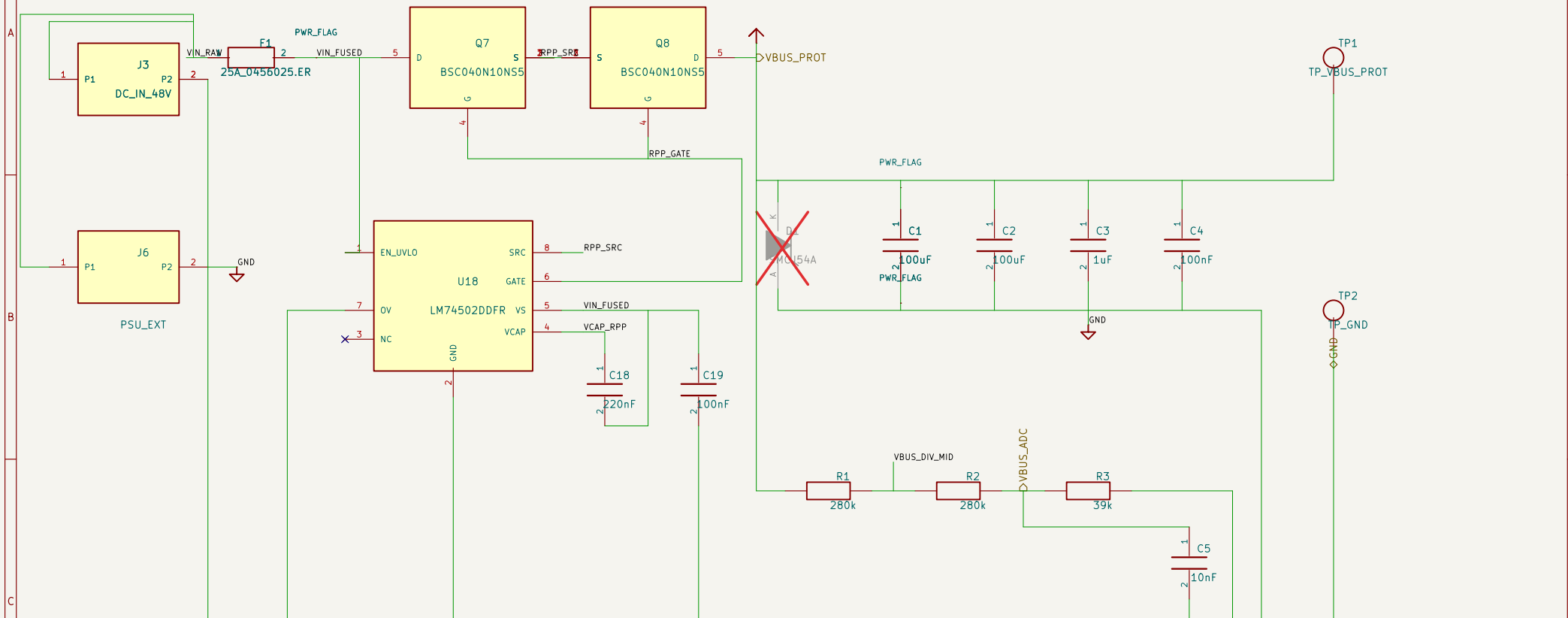


SYSTEM ARCHITECTURE – POWER FLOW / SAFETY FLOW / REAL-TIME I/O



SOURCE + -> J3.1 -> ONBOARD F1 -> LM74502 RPP -> DC-LINK



Q7/Q8 are back-to-back N-MOSFETs driven by LM74502 for low-loss reverse-polarity protection. Reverse current is not blocked when enabled; regeneration needs a receptive source. LM74502 OV is tied low in Rev.A1; positive surge is NOT claimed as closed. The 48V source/cable transient must be measured and kept inside the controller/downstream qualification envelope. F1: ONBOARD 0456025.ER 25A SMD. Verify fault current, inrush and thermal coordination. D1 stays DNP pending surge qualification.

VBUS_ADC divider = $280k + 280k \text{ over } 39k$, plus 10nF at the ADC node. Keep the high-side resistors split for voltage stress and route the sense return away from commutation current.
Local wiring: power, gate, divider/filter and ground. See docs/schematic_full_review_2026-10-09.md.

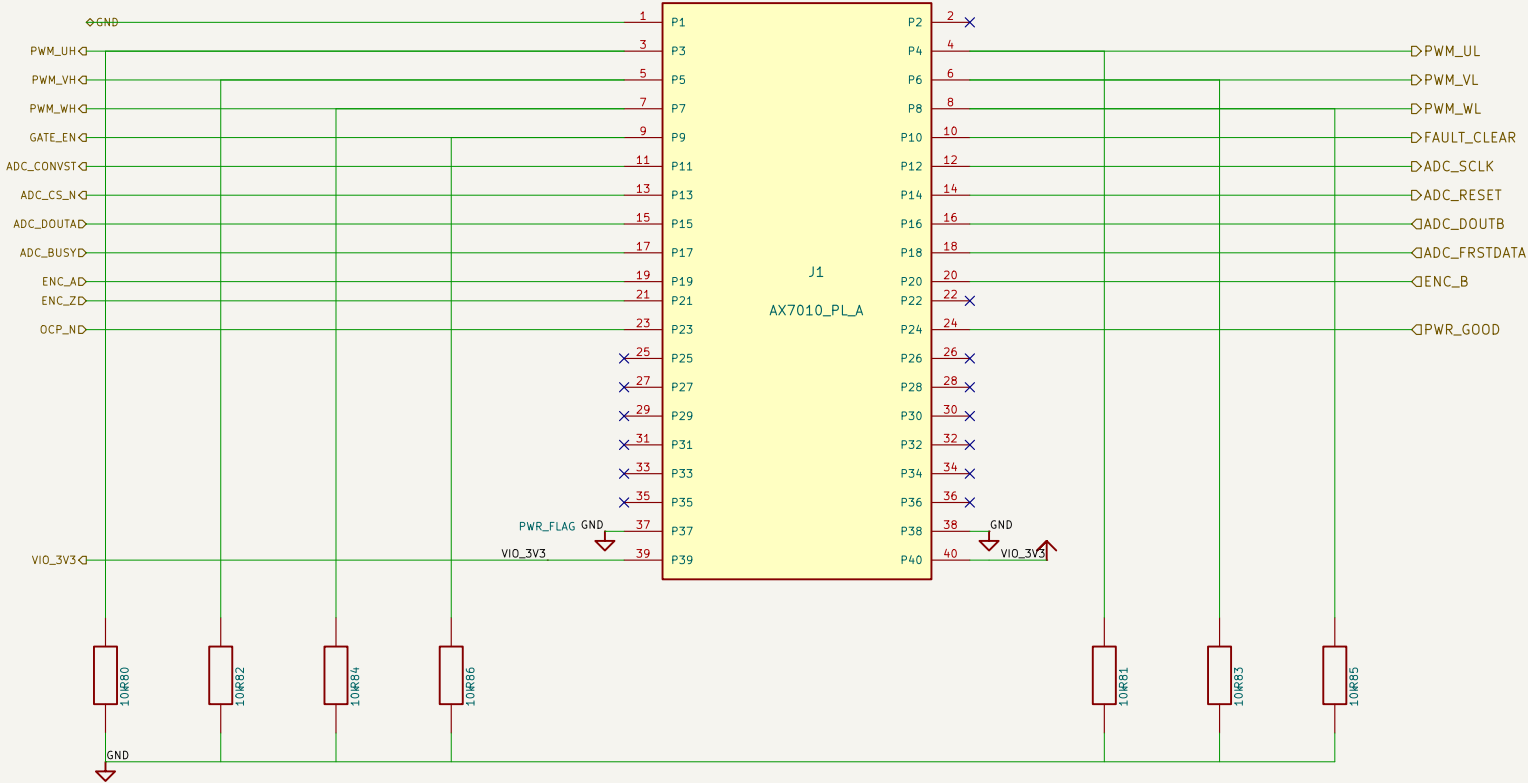
24..48V operating bus; source transient / regeneration qualification OPEN
magic-alt/fpga-servo
Sheet: /DC Input & Protection/
File: power_input.kicad_sch

Title: Rev.A1 - DC Input & Protection

Size: A4	Date: 2026-09-29	Rev: A1
KiCad E.D.A. 10.0.3		Id: 2/8

ALINX AX7010 (2022): active J10 interface + physical reset-state pull-down branches

J1 = AX7010 J10



R80..R86: each PWM and GATE_EN line has its own 10k pull-down.

J10 pin 2 (+5 V): NC. J2 removed; only J1 interfaces to AX7010.

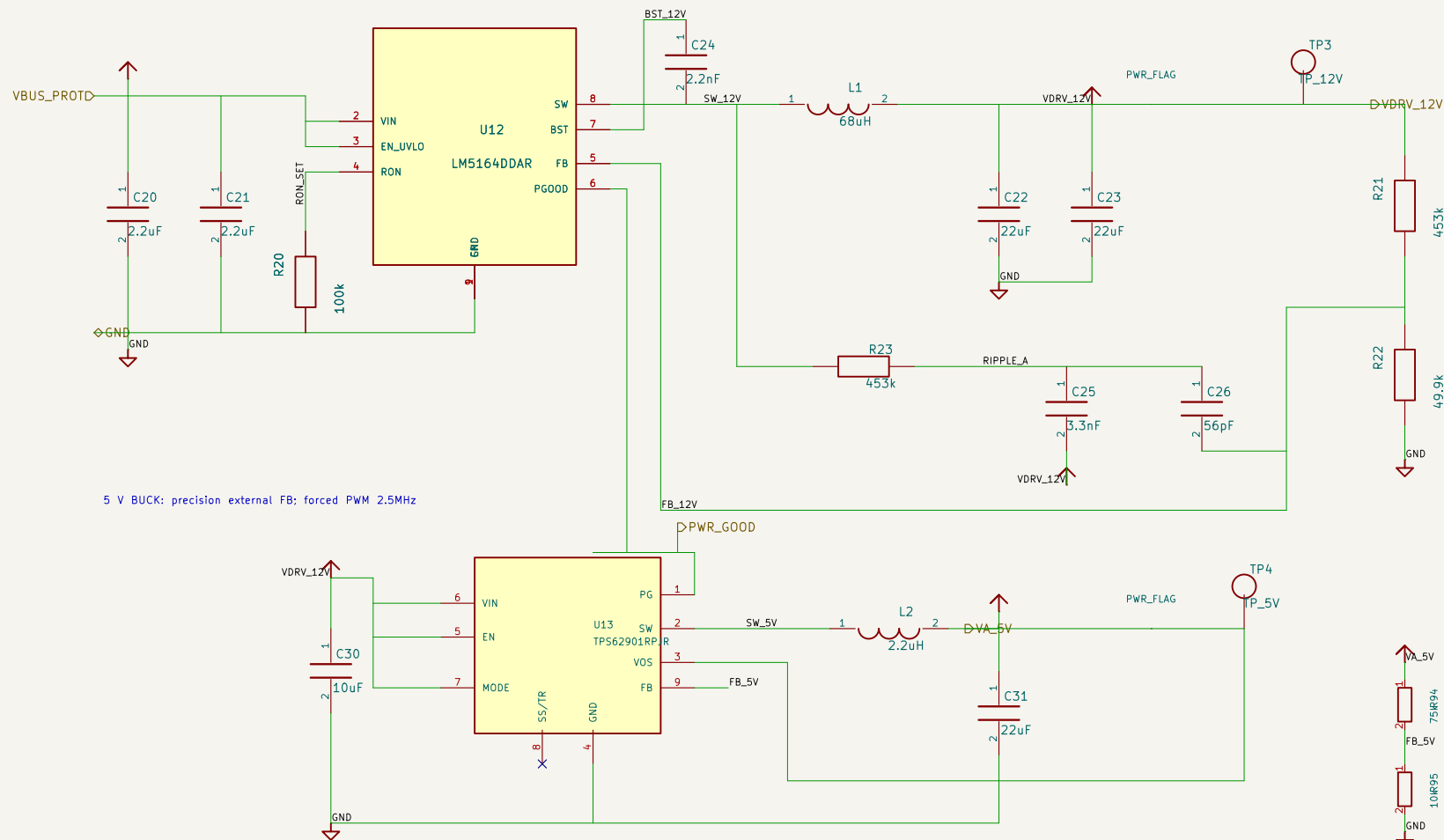
One 2x20 PL header J1 -> AX7010 J10; unused J2 removed
magic-alt/fpga-servo
Sheet: /AX7010 PL Interface/
File: ax7010_interface.kicad_sch

Title: Rev.A1 - AX7010 PL Interface

Size: A4	Date: 2026-09-29	Rev: A1
KiCad E.D.A. 10.0.3		Id: 3/8

VBUS_PROT -> LM5164 12V -> TPS62901 5.1V. Rail load / transient qualification OPEN.

12 V BUCK: input / bootstrap / output LC / feedback + ripple injection



5 V BUCK: precision external FB; forced PWM 2.5MHz

LM5164 values follow TI Figure 7-1: 15..100V In, 12V/1A, 300kHz, 68uH, 2x2.2uF CIN, 2x22uF COUT, 453k/49.9k feedback.
 TPS62901: 5.1V nominal; FB=75k/10k; MODE=VIN; forced PWM 2.5MHz. VOS to output.
 LM5164 PGOOD and TPS62901 PG share PWR_GOOD; pull up at 3.3V safety logic.
 System operating Input: 24..48V. Low-voltage bring-up is separately qualified.
 PGOOD alone does not prove VA5 valid; U25 separately monitors VA5.

LM5164 48V->12V + TPS62901 precision 5.1V

magic-alt/fpga-servo

Sheet: /Auxiliary Power/

File: aux_power.kicad_sch

Title: Rev.A1 - Auxiliary Power

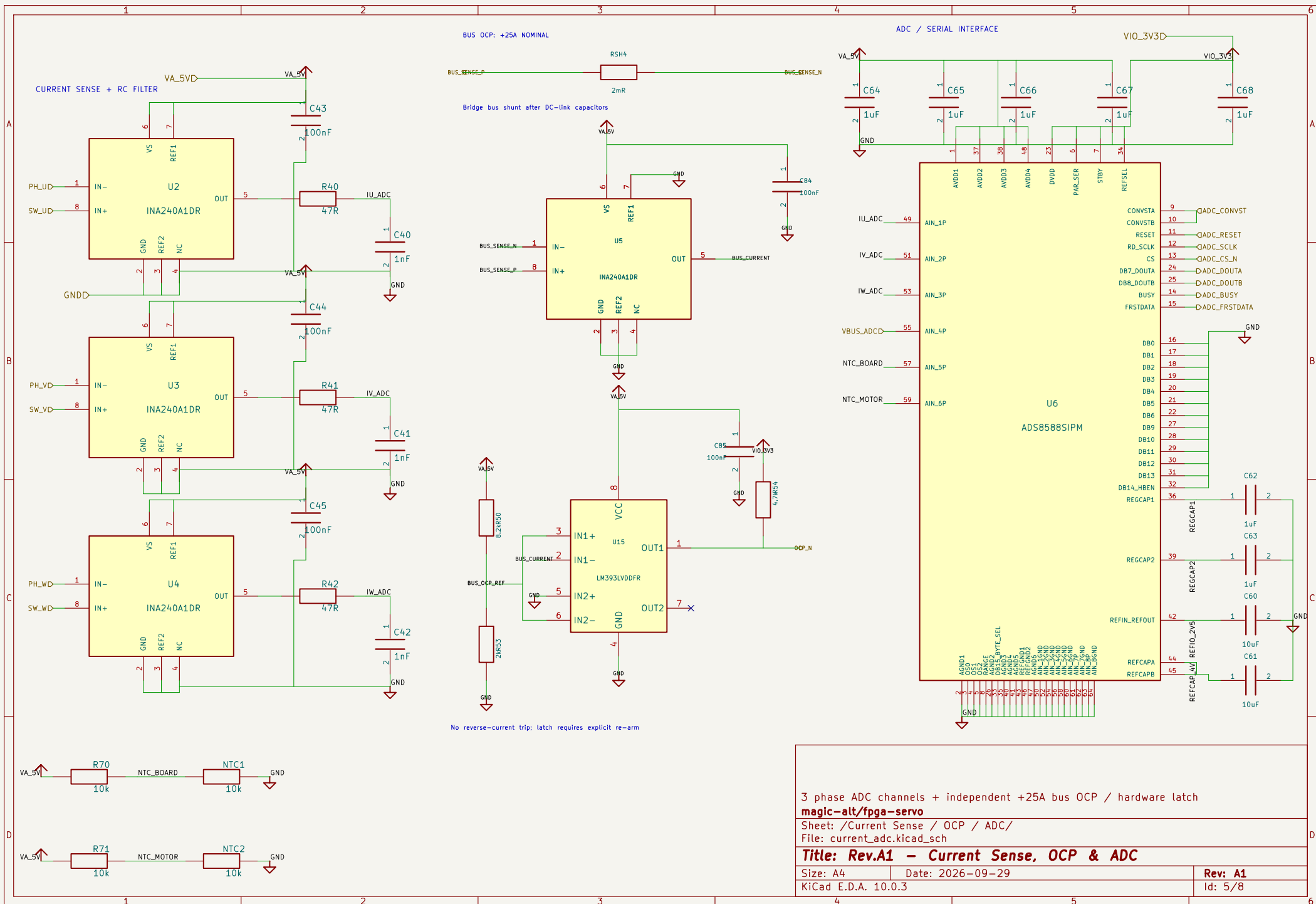
Size: A4

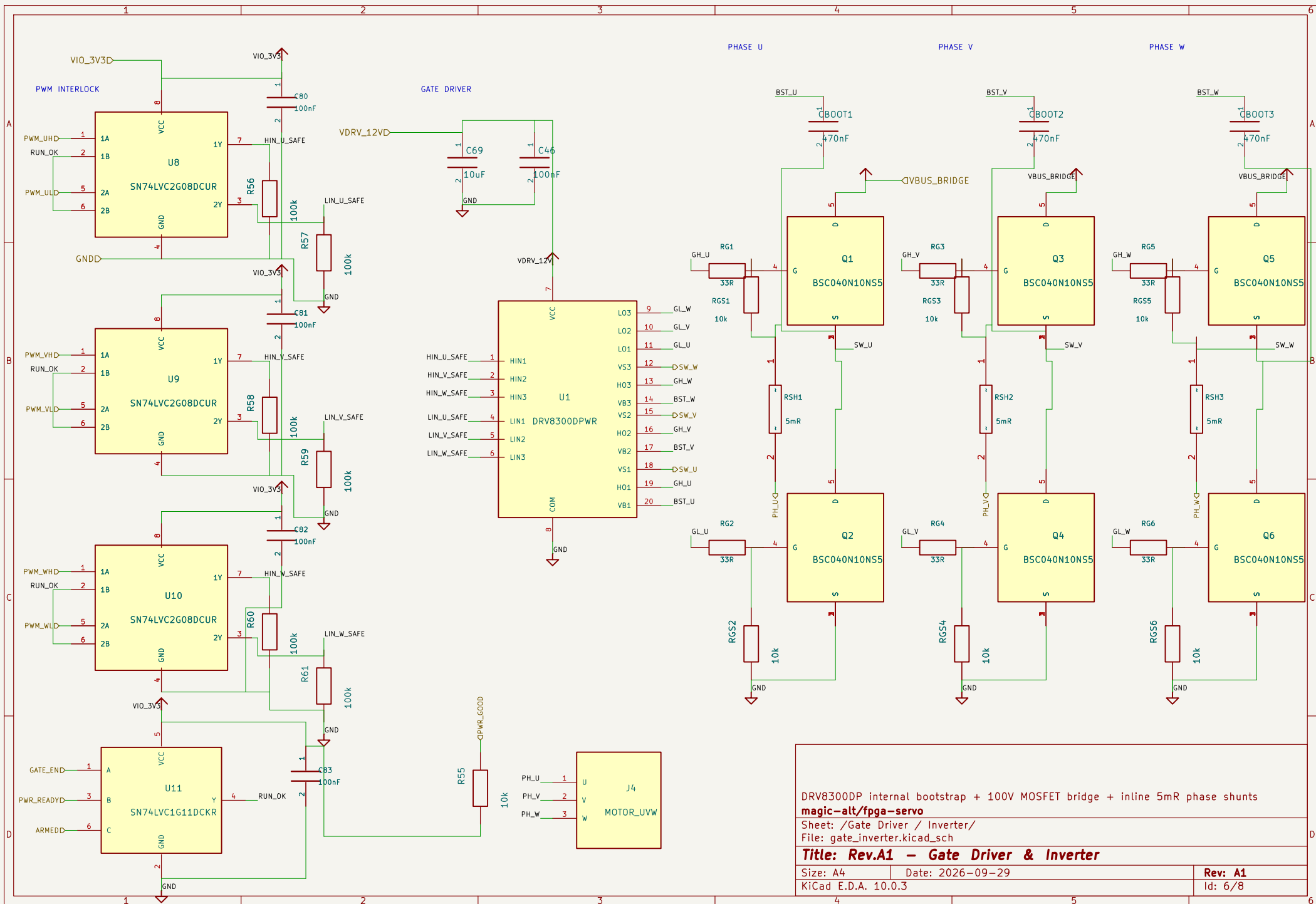
Date: 2026-09-29

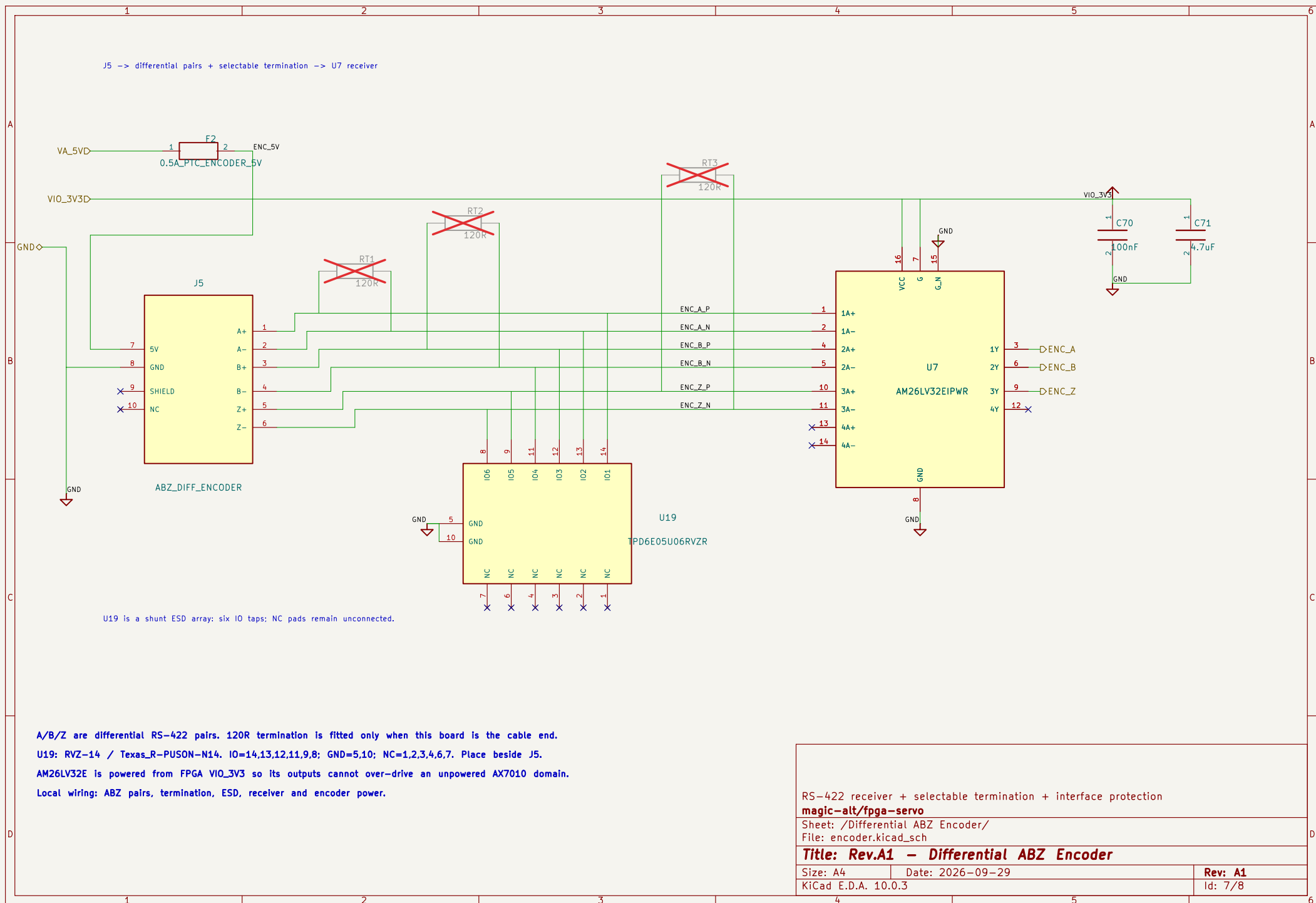
KiCad E.D.A. 10.0.3

Rev: A1

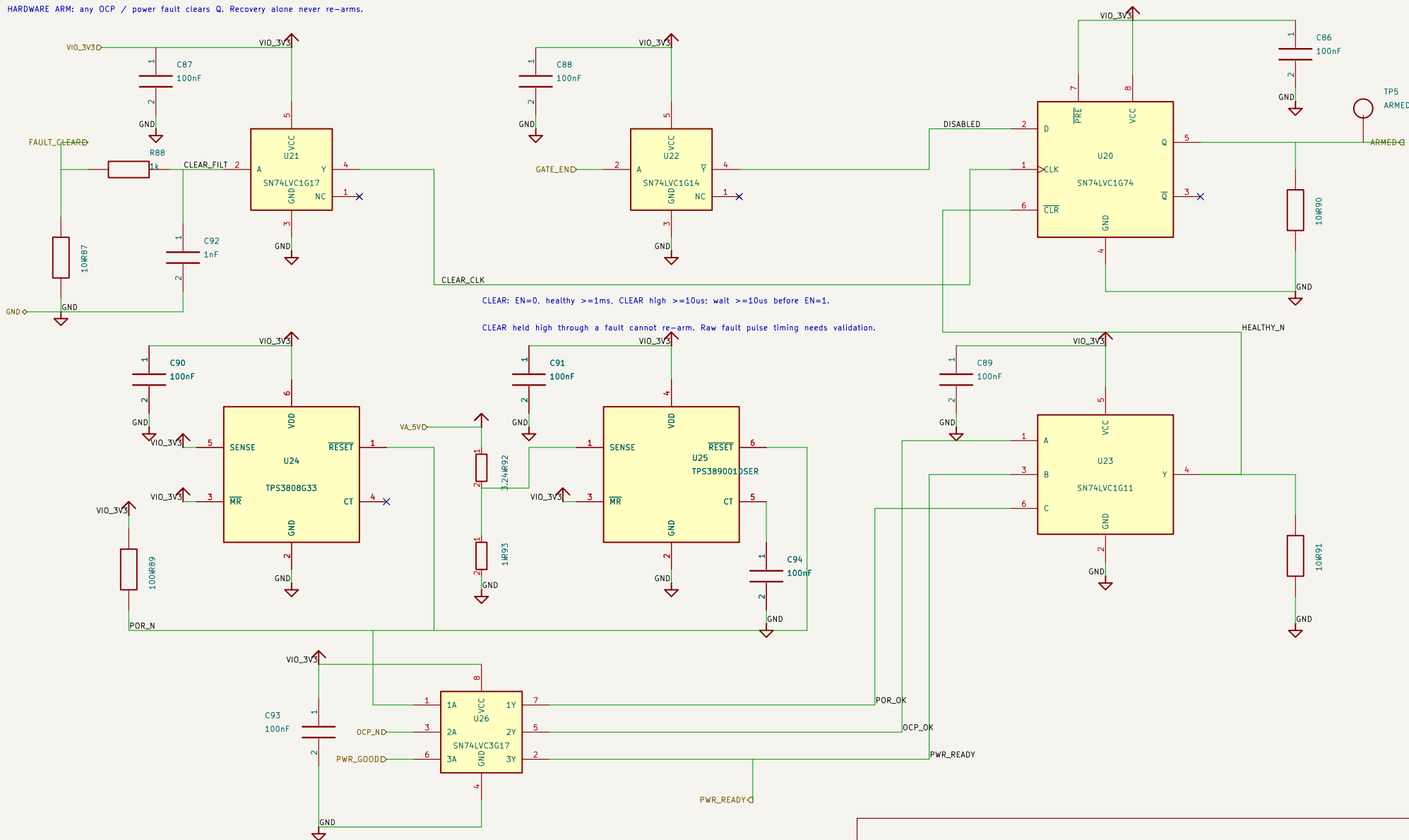
Id: 4/8







HARDWARE ARM: any OCP / power fault clears Q. Recovery alone never re-arms.



U24 CT open: 12..28ms. U25 CT=100nF: ~107ms; VA5 trip nominal 4.876V. Dynamic qualification OPEN.

U20..U26: local 100nF bypass; reset release passes through U26 Schmitt buffers.

magic-alt/fpga-servo

Sheet: /OCP hardware latch/

File: ocp_latch.kicad_sch

Title: Hardware OCP latch / supervised re-arm

Size: A4

Date: 2026-10-09

Rev: A1

KiCad E.D.A. 10.0.3

Id: 8/8